

Forecasting with hmmTMB: energy prices

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This vignette briefly showcases the `Forecast` class on the energy data set from the package `MSwM` (Sanchez-Espigares and Lopez-Moreno (2021)), which is also analysed in the main workflow vignette. For details about the variables, please refer to `?MSwM::energy`.

We start by loading a few packages and the energy data set.

```
# Load packages and data
library(hmmTMB)
library(ggplot2)
theme_set(theme_bw())
library(MSwM)
data("energy")

head(energy)
```

	Price	Oil	Gas	Coal	EurDol	Ibex35	Demand
1	3.188083	22.43277	14.40099	38.35157	1.134687	8.3976	477.3856
2	4.953667	22.27263	19.02747	38.35157	1.106439	8.3771	609.1261
3	4.730917	22.65383	18.48417	38.35157	1.106684	8.5547	650.3715
4	4.531000	23.67657	18.30143	38.35157	1.116819	8.4631	647.0499
5	5.141875	23.67209	14.55602	38.35157	1.122965	8.1773	627.9698
6	6.322083	23.60534	15.22485	38.35157	1.122460	8.1866	693.2467

Our aim will be to forecast the value of the `Price` variable (average price of energy). To compare the forecasted values to the truth, we split the data set into a training data set `data1`, which will be used to estimate the model parameters, and a forecast data set `data2`, which will only be used to provide covariates for forecasting. We will forecast over the final 100 days of the data set.

```
# Split data into training and test sets
energy$time <- 1:nrow(energy)
data1 <- energy[1:(nrow(energy)-100),]
data2 <- energy[(nrow(energy)-99):nrow(energy),]
```

We fit a 2-state hidden Markov model, where the response `Price` variable is modelled by a normal distribution in each state. The mean of the observation distribution is modelled as a function of the `Oil` and `Gas` variables, so that we can use future oil and gas prices to forecast energy prices.

```
# Hidden state model
hid <- MarkovChain$new(data = data1,
                      n_states = 2,
                      initial_state = 2)

# Observation process model
dists <- list(Price = "norm")
f <- list(Price = list(mean = ~Oil + Gas))
par0 <- list(Price = list(mean = c(2, 4), sd = c(2, 5)))
obs <- Observation$new(data = data1, dists = dists, formulas = f, par = par0)

# Fit HMM on training data
hmm <- HMM$new(obs = obs, hid = hid)
hmm$fit(silent = TRUE)
```

We use the `Forecast` class to compute forecast distributions for `Price`. `Forecast$new()` takes as inputs the fitted model object, a data set that includes covariates for the forecast times, and a list of grids of observation variable values at which to evaluate the forecast distributions. Based on the observed range of `Price` values, we choose a grid between 0 and 30.

```
# Define forecasting grid for Price distribution
obs_range <- list(Price = seq(0, 30, length = 1000))

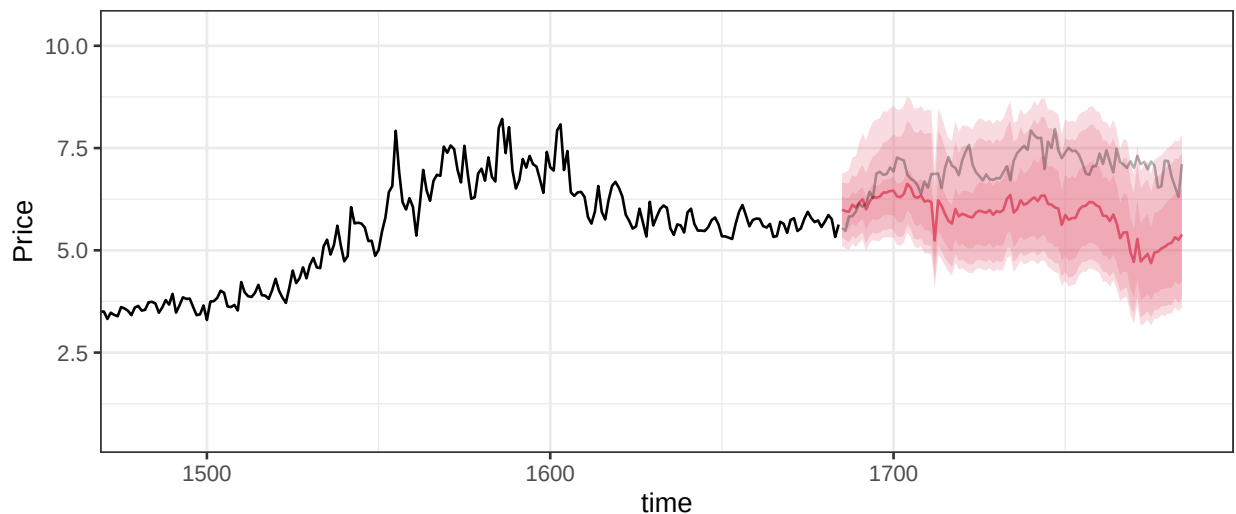
# Compute forecast distributions for test data
fc <- Forecast$new(hmm = hmm,
                  forecast_data = data2,
                  preset_eval_range = obs_range)
```

```
# Get forecast distribution means
head(fc$mean(var = "Price"))
```

```
[1] 5.992064 5.956509 5.940326 6.114813 6.044223 6.167915
```

We plot the forecast distributions with the `Forecast$plot()` method. This function shows the mean of the forecast distribution over time, as well as 50%, 80%, and 90% quantile forecast intervals (with decreasing opacity). We use `show_data = TRUE` to include the training data (before the forecasting period), and add a line for the true Price values during the forecast period for comparison.

```
# Plot mean and quantiles of forecast distributions
fc$plot(var = "Price", show_data = TRUE) +
  coord_cartesian(xlim = c(nrow(data1) - 200, nrow(data1) + nrow(data2))) +
  geom_line(aes(y = Price), data = data2, alpha = 0.3)
```



References

Sanchez-Espigares, Josep A., and Alberto Lopez-Moreno. 2021. *MSwM: Fitting Markov Switching Models*. <https://CRAN.R-project.org/package=MSwM>.